

HMQ: Hardware Friendly Mixed Precision Quantization Block for CNNs

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Abstract

Recent work in network quantization produced state-of-the-art results using mixed precision quantization. An imperative requirement for many efficient edge device hardware implementations is that their quantizers are uniform and with power-of-two thresholds. In this work, we introduce the Hardware Friendly Mixed Precision Quantization Block (HMQ) in order to meet this requirement. The HMQ is a mixed precision quantization block that repurposes the Gumbel-Softmax estimator into a smooth estimator of a pair of quantization parameters, namely, bit-width and threshold. HMQs use this to search over a finite space of quantization schemes. Empirically, we apply HMQs to quantize classification models trained on CIFAR10 and ImageNet. For ImageNet, we quantize four different architectures and show that, in spite of the added restrictions to our quantization scheme, we achieve competitive and, in some cases, state-of-the-art results.

1 Introduction

In recent years, convolutional neural networks (CNNs) produced state-of-the-art results in many computer vision tasks including image classification [14, 17, 21, 22, 38, 39], object detection [29, 36, 40], semantic segmentation [31, 37], etc. Deploying these models on embedded devices is a challenging task due to limitations on available memory, computational power and power consumption. Many works address these issues using different methods. These include pruning [16, 45, 47], efficient neural architecture design [14, 21, 24, 38], hardware and CNN co-design [14, 20, 43] and quantization [6, 13, 15, 23, 24, 46].

In this work, we focus on quantization, an approach in which the model is compressed by reducing the bit-widths of weights and activations. Besides re-

The code of this work is available in <https://github.com/sony-si/ai-research>.